

# Number System

1 Octal = 8 bits.

## Decimal to Binary Conversation.

| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |
|-----|----|----|----|---|---|---|---|
|-----|----|----|----|---|---|---|---|

0 1 0 0 1 1 0 0 0 → 74

0 1 0 1 0 0 1 0 → 82

0 1 1 0 0 1 0 0 → 100

1 0 0 1 0 1 0 1 → 150

1 0 0 0 0 0 0 0 → 128

1 1 1 0 0 0 0 0 → 224

1 1 0 1 0 0 0 1 → 241

1 1 0 1 1 0 1 0 → 250

1 1 1 1 1 0 1 0 → 255

0 1 0 1 0 1 0 1 → 21

1 1 1 1 1 0 0 0 → 248

1 1 1 0 1 0 1 1 → 255

1 0 1 1 0 0 0 0 → 90

0 1 0 1 0 0 0 0 → 192

1 1 1 1 1 1 1 1 → 255

1 1 1 1 1 1 1 1 → 255

1 1 1 1 1 1 1 1 → 255

1 1 1 1 1 1 1 1 → 255

1 1 1 1 1 1 1 1 → 255

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |
|-----|----|----|----|---|---|---|---|

1 1 1 1 0 0 0 0 → 240

0 1 1 1 0 1 0 0 → 120

2 1 1 1 1 0 1 → 254

Hexa to Binary Conversion:

1 hexa = 4 bits.

|   |   |   |   |
|---|---|---|---|
| 8 | 4 | 2 | 1 |
|---|---|---|---|

A → 1 0 1 0

8AF → 1 0 0 0 1 0 1 0 1 1 1 1

1C0 → 0 0 0 1 1 1 0 0 0 0 0 0

FF0 → 1 1 1 1 1 1 1 1 0 0 0 0

AAA → 1 0 1 0 1 0 1 0 1 0 1 0

ABC → 1 0 1 0 1 0 1 1 1 1 0 0

42D → 0 1 0 0 0 0 1 0 1 1 0 1

2D0 → 0 0 1 0 1 1 0 1 0 0 0 0

FFF → 1 1 1 1 1 1 1 1 1 1 1 1

FAB → 1 1 1 1 1 0 1 0 1 0 1 1

CC0 → 1 1 0 0 1 1 0 0 0 0 0 0

F10  $\rightarrow$  1111 0001 0000

221  $\rightarrow$  0010 0010 0001

CB5  $\rightarrow$  1100 1011 0101

48F  $\rightarrow$  0100 1000 1111

99D  $\rightarrow$  1001 1001 1101